MagicCode Super-Resolution Software Getting Started

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1. Unified Architecture

MagicSR v2.1.0 provides a single static library per platform and a single header (interface/mc_interface.h). All super-resolution operations use MC_Enable and MC_Disable.

Platform	Static Library	Header
Android	lib/android/libmagic_sr.a	interface/mc_interface.h
iOS	lib/ios/libmagic_sr.a	interface/mc_interface.h
macOS (Apple Silicon)	lib/mac_arm/libmagic_sr.a	interface/mc_interface.h
macOS (x86_64)	lib/mac_x86/libmagic_sr.a	interface/mc_interface.h
Windows	lib/windows/libmagic_sr.lib	interface/mc_interface.h

2. Quick Start Example

- Include interface/mc_interface.h and link libmagic_sr.a (or libmagic_sr.lib).
- Declare void *handle = NULL.
- Populate input_param_t and magic_frame_t.
- Call MC_Enable(&handle, &frame, &param, &status).
- On subsequent frames with unchanged parameters, call MC_Enable(&handle, &frame, NULL, &status).
- When finished, call MC_Disable(handle).

Temporal Super-Resolution

v2.1.0 provides temporal super-resolution through the unified public C ABI: libmagic_sr.a (Windows: libmagic_sr.lib) + interface/mc_interface.h + MC_Enable / MC_Disable. A single function, MC_Enable, handles implicit handle initialization, dynamic reconfiguration, temporal execution, and execution status query.

Modes

Value	Enumerator	Role
0	SPATIAL_SPEED_MODE	Spatial, throughput first
1	SPATIAL_BALANCED_MODE	Spatial, quality/cost trade-off
2	TEMPORAL_SPEED_MODE	Temporal, throughput first
3	TEMPORAL_BALANCED_MODE	Temporal, canonical FSR-family path

- Temporal scaler_factor is (1, 8]; 1.0 itself is rejected. Spatial remains [1.0, 8.0].
- TEMPORAL_SPEED_MODE on Metal, Vulkan, and OpenGLES currently uses a lightweight Convert → Activate → Upscale pipeline. Direct3D 11 and desktop OpenGL 4.3 Speed use the FSR-family temporal path (same family as Balanced, not a separate customer API).
- TEMPORAL_BALANCED_MODE uses the canonical FSR-family path on every GPU temporal backend. Desktop Balanced (Windows Vulkan / Direct3D 11 / desktop GL; Apple Metal) can run edge-resolve; if enable_sharpening is on, RCAS runs after edge-resolve.
- CPU backends (X86 / NEON) cannot run temporal modes.
- The API accepts (1, 8]. Exact 2× has optimized kernels. Not every scale or device is claimed as device-run.
- Internal pipeline names are implementation detail, not public configuration.

Minimum per-frame inputs

- magic_frame_t.image_in / image_out: caller-owned color and output GPU resources.
- magic_frame_t.frame: non-NULL temporal_frame_t* with struct_size = sizeof(temporal_frame_t).
- temporal_frame_t.depth and motion at input resolution.
- jitter_offset_x/y in input pixels, top-left, +X right, +Y down.
- frame_index / reset_history; camera_near / camera_far / camera_fov_y (radians).
- gpu_context at initial MC_Enable: Vulkan requires physical_device + device; D3D11 requires device; Metal device may be NULL (system default); GL/GLES use the current context (library never makes current).
- Vulkan temporal requires magic_frame_t.command_buffer to be a recording VkCommandBuffer. Metal command_buffer is optional. D3D11 / GL / GLES leave it NULL.
- Create-stage depth_reversed / depth_infinite / hdr_color on input_param_t (all-zero = conventional-Z, finite far, LDR). Immutable for the handle lifetime.

Motion, depth, and masks

- Motion is current → previous. The library never negates MVs.
- motion_vector_scale_x/y is the only MV numeric conversion. (0,0) defaults to (input_width, input_height). Mixed zero is rejected.
- Depth is GPU device depth in [0, 1]. Linear view-space depth is not accepted.
- reactive / transparency are optional. Explicit caller textures win per channel.
- When omitted, the library derives an approximate mask. That is not a semantic transparency mask.
- Speed on Metal/Vulkan/GLES may derive both missing reactive and an approximate transparency. FSR-family paths (Balanced on all backends; Speed on D3D11/desktop GL) derive missing reactive only and bind an internal zero transparency texture.
- Both temporal modes enable depth+MV history rejection by default.
- Diagnostic only: MAGICSR_TAAU_AUTO_MASK=0 and MAGICSR_TAAU_HIST_REJECT=0 can turn those defaults off. They are not a customer configuration path.

3. Models

Models are located in model/ and deployed via tools/setup_models.sh. Specify the path to model files in input_param_t.model_path.

4. Next documents

- User Guide: doc/User_Guide.md and doc/用户使用说明书.md
- API Reference, Integration Guide, FAQ, Migration Guide, Known Issues in this folder